



Fuel Control law for engine Playbook

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Step 1: System Modeling

- **Define the System:**
Water injection is modeled as an additional flow input into the combustor that reduces flame temperature by increasing specific heat capacity and evaporation.
- **State-Space Representation:**
States: combustor pressure, temperature, species mass fractions.
Inputs: fuel mass flow, water mass flow, compressor exit pressure (P3).
Outputs: turbine inlet temperature (TIT), emissions (NO_x).
- **Linearization:**
Linearize around takeoff/cruise operating points for small perturbations of water/fuel flows.

Step 2: Define Control Objectives

- **Performance Requirements:**
Maintain TIT below limits, reduce NO_x emissions during takeoff, preserve combustion stability.
- **Setpoint Tracking:**
Track TIT and EGT targets while respecting thrust demand.
- **Disturbance Rejection:**
Handle variations in fuel quality, ambient humidity, inlet conditions.
- **Constraints:**
 - Actuator limits: maximum water flow rate.



- State limits: flame stability margin, maximum EGT.

Step 3: Control Law Design

- **Classical Control:**
PID loop for TIT regulation via water injection rate.
- **Modern Control:**
MPC for minimizing TIT and NOx while maintaining thrust.
- **Robust Control:**
 H^∞ design for uncertain combustor dynamics.
- **Adaptive/Intelligent Control:**
Adaptive gain tuning for varying altitudes and inlet conditions.

Step 4: Simulation and Analysis

- **Build Simulation:** MATLAB/Simulink model with combustor + water injection actuator dynamics.
- **Test Cases:**
 1. Takeoff (high P3, high TIT, water injection ON).
 2. Cruise (steady-state, minimal water injection).
 3. Transient throttle increase (verify TIT limiting).
- **Evaluation Metrics:** TIT tracking error, NOx reduction %, combustion stability index.

Abstract

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Aircraft gas turbine engines rely on sophisticated fuel control logics to ensure safe starting, transient operability, efficiency, and emissions compliance across wide flight envelopes. Core strategies include start sequencing, N-based scheduling, fuel pumping and metering, acceleration/deceleration limiting, flame stability management, emission minimization, and cascaded spool-speed (NL, NH) loops. This paper formulates the governing equations, links them to control architectures (FADEC), and cites NASA and SAE literature on implementation and verification.

1. Start Sequencing

Purpose: Initiate engine from rest to idle, avoiding surge or overtemperature.

Logic:

$$W_f(t) \leq W_{f,\max}(T_{t4}, N_2)$$

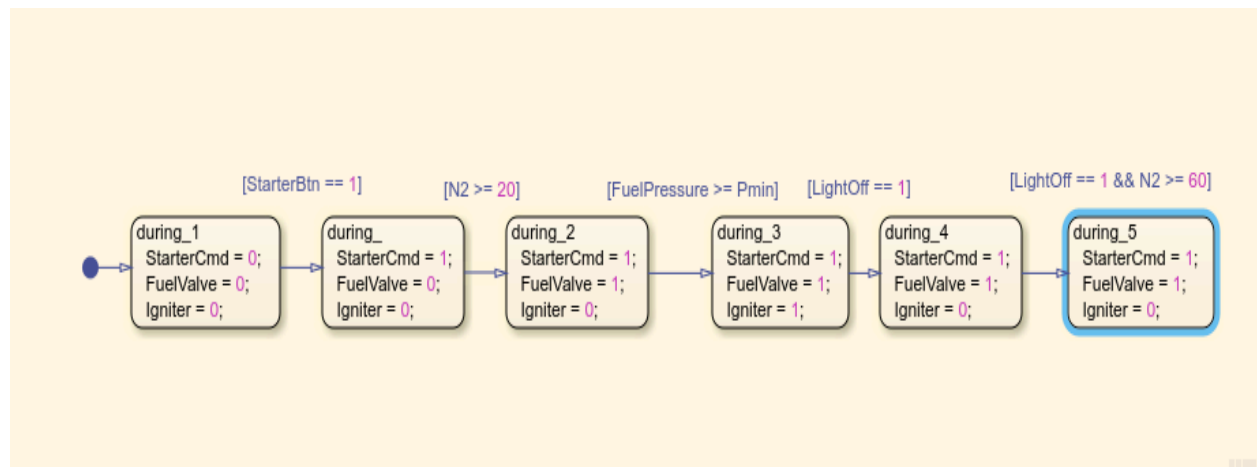
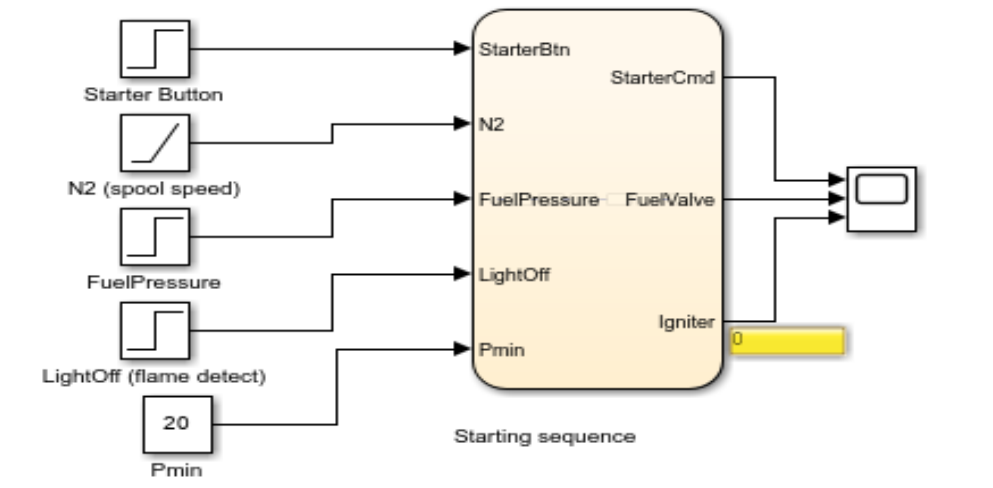
$W_f(t)$ – Instantaneous commanded fuel flow rate [kg/s].

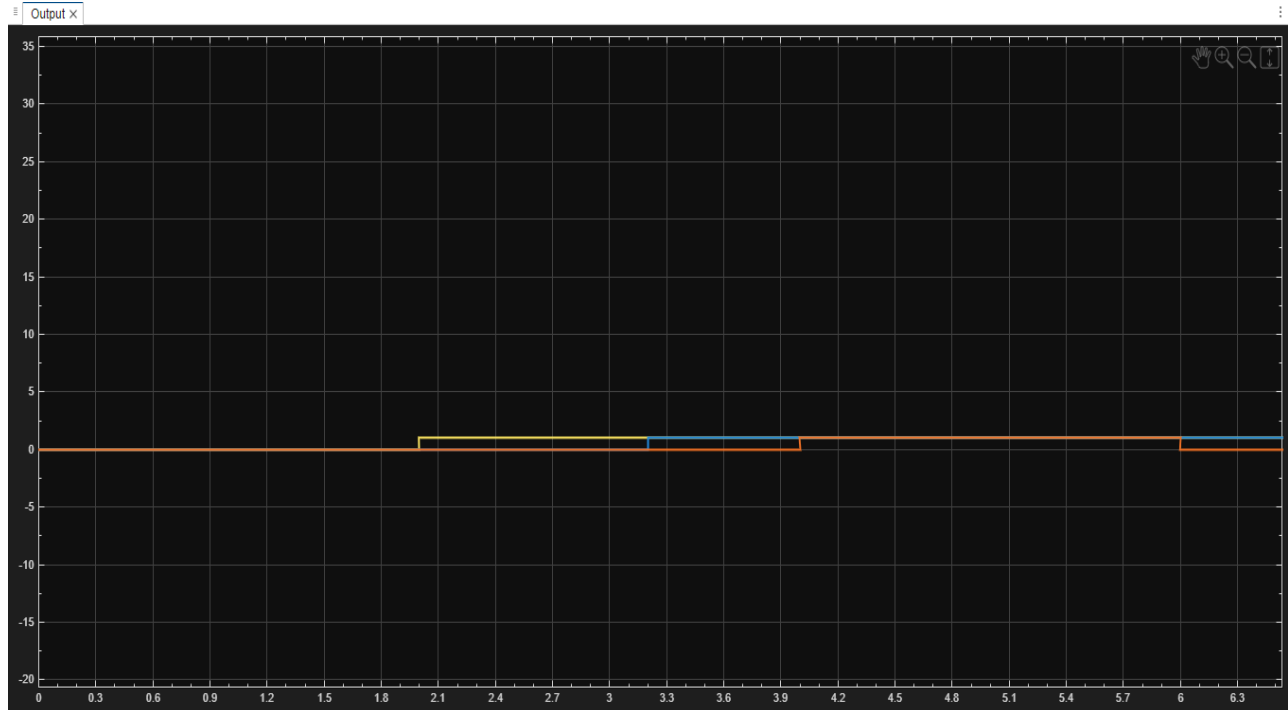
$W_{f,\max}$ – Maximum allowable fuel flow as a function of combustor temperature and spool speed.

T_{t4} – Combustor outlet total temperature (station 4), also called turbine inlet temperature.

N_2 – High-pressure spool (HP compressor shaft) rotational speed [% or rpm].

Fuel is enabled after the HP spool reaches ~20% N_2 ; starter cut once self-sustaining. Transient fuel ramp optimization has been developed under NASA's **TTECTra** tool.





The developed Stateflow model represents the **engine starting sequence** by controlling three main outputs: **Starter Command, Fuel Valve, and Igniter**. The sequence is designed such that each output activates at a different stage, allowing the engine to transition smoothly from idle to self-sustaining operation.

From the simulation results, the output graph shows a **step-like progression**:

1. At around **2 seconds**, the **Starter Command** activates, initiating engine spool-up [**Yellow**].
2. At around **3.2 seconds**, once conditions such as sufficient spool speed (N2) and fuel pressure are satisfied, the **Fuel Valve** opens [**Blue**].
3. At around **4 seconds**, the **Igniter** turns ON, ensuring combustion initiation [**Orange**].

This staggered activation produces a **clear staircase pattern** in the Scope output, with each signal rising at different times. The model demonstrates how logic-based conditions can be used to coordinate the starter, fuel, and ignition systems during engine start-up.



2. Fuel Schedule Definition – N Control

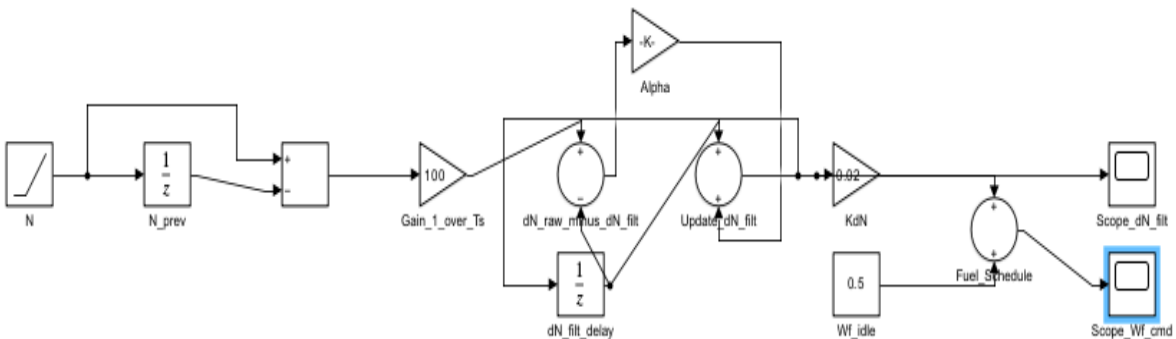
Purpose: Limit commanded fuel by spool acceleration.

$$W_{f,cmd} = K_{\dot{N}} \cdot \frac{dN}{dt} + W_{f,idle}$$

Where:

- $W_{f,cmd}$ = commanded fuel flow
- $K_{\dot{N}}$ = gain relating spool acceleration to fuel demand
- $\frac{dN}{dt}$ = rate of change of spool speed
- $W_{f,idle}$ = fuel flow at idle (minimum flow baseline)

This ensures the fuel system only allows as much extra fuel as the engine can safely accelerate with.

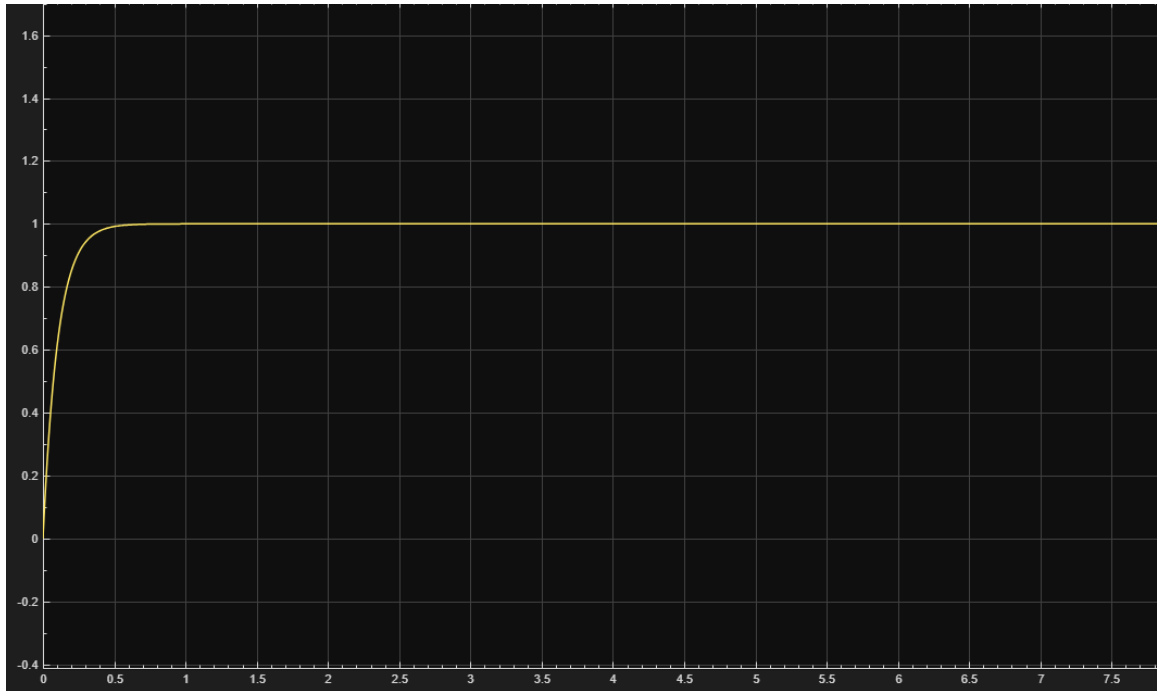




Block Diagram Description

The Simulink implementation of the fuel scheduling logic consists of the following elements:

1. **Ramp Block (Input N)** – Generates a gradual increase in engine shaft speed to simulate acceleration.
2. **Derivative Block** – Computes N' , the rate of change of shaft speed.
3. **First-Order Low-Pass Filter** – Processes N' to obtain a smooth derivative signal ($dN_{filt}dN$), avoiding abrupt changes.
4. **Gain Block (KdN)** – Scales the filtered derivative according to the desired sensitivity of fuel flow to speed changes.
5. **Constant Block (Wf_idle)** – Provides the idle baseline fuel flow.
6. **Sum Block** – Adds the idle flow with the scaled derivative contribution to generate the commanded fuel flow Wf_{cmd} .
7. **Scopes** –
 - *Scope_ dN_{filt}* : Displays the filtered derivative response, which rises and settles at unity for a unit slope input.
 - *Scope_ Wf_{cmd}* : Displays the commanded fuel flow, which starts at 0.5 and gradually settles at 0.52.



In the simulation model, both outputs exhibit a gradual rise and settling behavior due to the combined effects of the ramp input, derivative calculation, and filtering dynamics. The input N is defined as a ramp with a slope of 1, meaning its true derivative is a constant value of 1. However, instead of computing the derivative directly, the model applies a discrete low-pass filter. This filter smooths the derivative signal, causing the filtered derivative output (dN_filt) to start from zero, rise gradually, and asymptotically approach the steady-state value of 1.

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The commanded fuel flow (Wf_cmd) is then scheduled based on this filtered derivative. It is defined as the sum of the idle fuel flow baseline ($Wf_idle = 0.5$) and the scaled contribution of the derivative term ($KdN \cdot dN_{filt}$). With $KdN = 0.02$, the final steady-state value becomes $0.5 + 0.02 \times 1 = 0.52$. As a result, the Wf_cmd output begins at the idle value of 0.5, increases smoothly following the filter dynamics, and settles at 0.52.

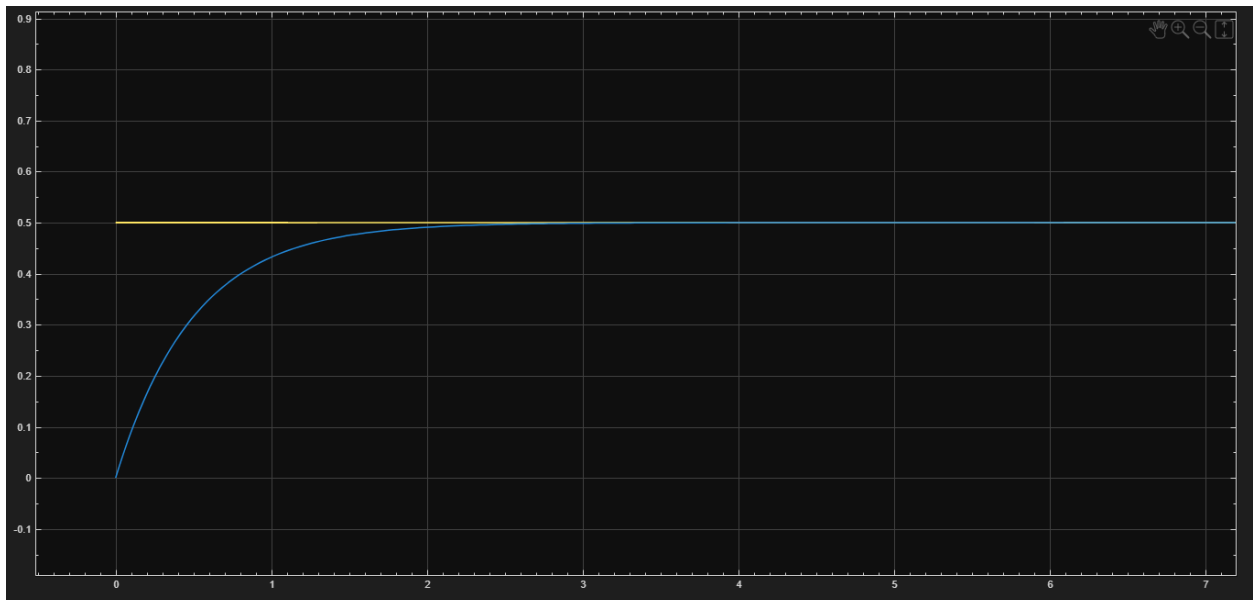
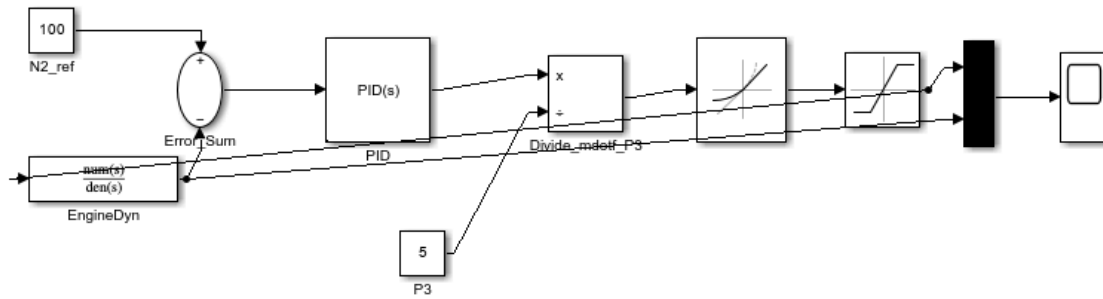
Thus, both outputs show exponential-like responses instead of instantaneous steps: the derivative output rises to unity, and the commanded fuel flow rises from idle to its scheduled value. This behavior demonstrates the filtering effect in smoothing transient responses and ensuring gradual fuel flow scheduling.

3. Fuel schedule definition - $\dot{m}_{dotf}/P3$ controller - setpoint controller

In modern gas turbine engines, **fuel flow scheduling** is a critical function of the Full Authority Digital Engine Control (FADEC). The goal is to regulate the high-pressure spool speed ($N2$) according to pilot demand while maintaining engine safety.

A commonly used approach normalizes the fuel flow $\dot{m}_{dotf}/p3$ by the compressor discharge pressure $P3$, since this improves robustness across varying operating conditions.

This document describes a Simulink implementation of a **$\dot{m}_{dotf}/P3$ controller with setpoint speed tracking**, along with the mathematical formulation.



Control Structure

The implemented control system consists of the following elements:

1. Setpoint Input (N2,ref)

A constant block providing the desired reference for the high-pressure spool speed.

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2. Engine Dynamics (First-Order Lag)

The engine response to commanded fuel flow is modeled by a first-order transfer function

$$\frac{N_2(s)}{W_f(s)} = \frac{K}{\tau s + 1}$$

where $K=1$ (normalized gain) and $\tau=0.5$ s

Error Calculation

The tracking error is defined as:

$$e(t) = N_{2,ref}(t) - N_2(t)$$

PID Controller

The controller generates a control effort based on:

$$u(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{de}{dt}$$

In the model, $K_p=0.05$, $K_i=0.01$, $K_d=0.001$

Fuel Scheduling via Normalization

The raw control effort is normalized by compressor discharge pressure P_3 :

$$W_{f,raw} = \frac{u}{P_3}$$

where $P_3=5$ (constant in this simulation).

Acceleration/Deceleration Limiter

To ensure safe transients, the fuel flow rate is rate-limited:



$$\frac{dW_f}{dt} \leq L_{acc}, \quad \frac{dW_f}{dt} \geq L_{dec}$$

with $L_{acc} = 0.05 \text{ kg/s}^2$, $L_{dec} = -0.03 \text{ kg/s}^2$.

Above Simulink model (fuel schedule + PID + divider $\dot{m}_f / P3$ + limiter + engine dynamics) and the **scope result** with two outputs plotted:

- **Yellow line (fuel flow command after schedule/limit):** flat at the upper bound (≈ 0.5).
- **Blue line (N_2 response):** starts low, rises smoothly, and settles near the commanded level.

Equation inside your model:

$$\dot{m}_{f,\text{cmd}} = \frac{PID(N_{2,\text{ref}} - N_2)}{P3}$$

with saturation and rate-limiting blocks to ensure flame stability and surge margin.

Result behavior:

- The fuel schedule limits the maximum commanded fuel - why yellow is flat at 0.5.
- The engine dynamics (1st-order lag) makes N_2 rise gradually toward equilibrium instead of instantly.

$$\frac{\dot{m}_f}{P_3} = f(PID(e_{N_2}))$$

where



- $\dot{m}_f$ = commanded fuel flow
- P3 = compressor discharge pressure
- $e_{N2} = N2_{ref} - N2_{actual}$ = error in core speed
- PID = controller regulating acceleration/deceleration

The **normalization of fuel flow by P3** is standard in **gas turbine control** because it provides a **dimensionless scheduling parameter**, robust across altitude/ambient pressure changes.



Fuel system control logics- Starting sequence

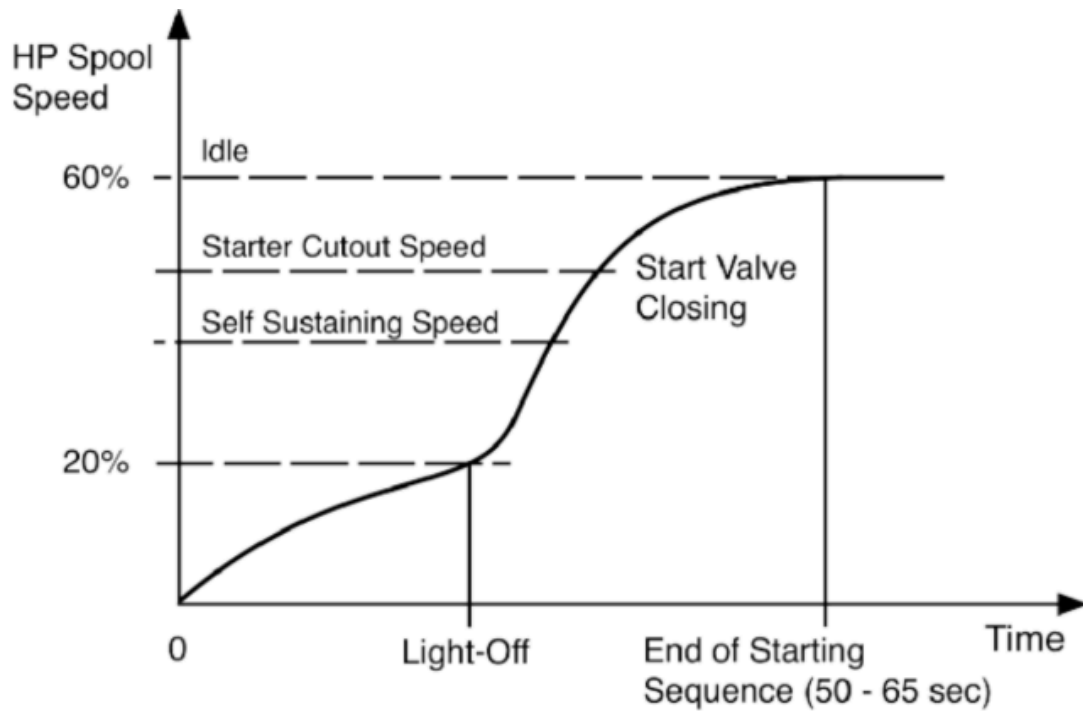
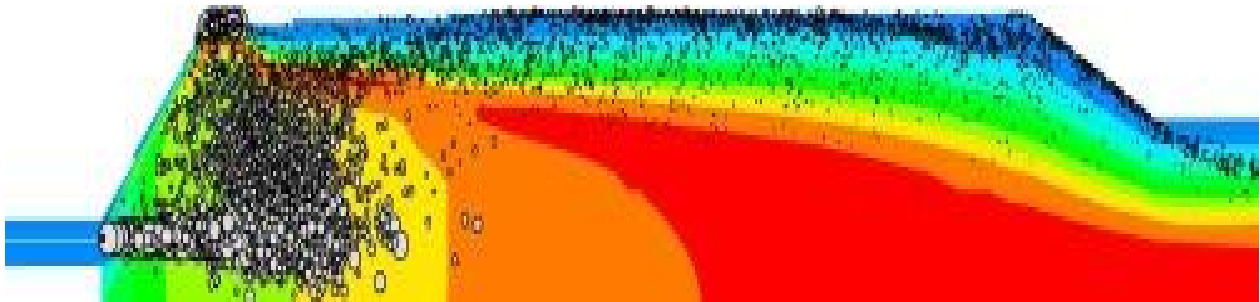


Fig. 7.1 Acceleration of the high pressure spool during the start sequence. The typical speeds that are important during starting are shown

Combustor — 1.3.1 Water Injection Control



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System Modeling

- Water injection $\rightarrow$ evaporation + heat capacity change $\rightarrow$ TIT reduction. Include injector dynamics and humidity coupling.

Control Objectives

- Reduce peak TIT/NOx during high thrust while preserving flame stability.

Control Law Design

- Feedforward schedule (PLA $\rightarrow$ Ww) + TIT feedback PI; safety interlocks for flame quench and max water limits.
- Optional MPC to trade TIT vs NOx under constraints.

Control Law Design

Control architecture (recommended, practical FADEC style)

1. Scheduler / Mode Logic (supervisory)

- Enable water injection only in modes that permit it (e.g., takeoff/climb/A/B on).
- Compute a feedforward water schedule $Ww_{ff} = f(PLA, Wf, P3, \text{altitude, ambient_temp})$ — a lookup table derived from test/CFD to meet target Tt4 reduction.

2. Primary Control Loop (feedback)

- **Outer loop:** PI controller on Tt4 error: $e = Tt4_{ref} - Tt4_{meas} \rightarrow$ PI produces ΔWw_{fb} .
- **Inner actuator handling:** rate limiter + saturation on commanded Ww.



- Command: $Ww_cmd = \text{saturate}(Ww_ff + \Delta Ww_fb)$.

3. Advanced options

- **MPC / constrained optimizer** at supervisory level: minimize NOx subject to Tt4 and actuator constraints when water is scarce or multi-objective tuning required.
- **Adaptive gain scheduling**: adjust PI gains with altitude/P3 and equivalence ratio.
- **Robustification**: H^∞ or loop shaping when plant uncertainty high (evaporation times vary).

4. Safety & Interlocks

- Flame stability monitor (chemiluminescence/pressure oscillations/combustor pressure): if flame stability index < threshold → reduce/disable Ww immediately.
- Low-water sensor → graceful disable + log/fault.
- Tt4 sensor failure: fallback to conservative schedule or disable injection.

Suggested controller form

- PI: $Ww_fb(s) = K_p (Tt4_err) + K_i \int Tt4_err dt$
- Anti-windup on integrator; derivative not recommended (sensor noise).

Simulation & Test Cases

- Takeoff transients, back-to-back high thrust cycles, water supply depletion.
- Metrics: TIT peaks, NOx proxy change, flame stability indicator.



Combustor — 1.3.2 Afterburning Control (if present)

System Modeling

- Afterburner fuel injection dynamics, residence time, ignition logic, pressure rise coupling to upstream systems.

Control Objectives

- Smooth ignition/extinguish, controlled thrust augmentation, avoid pressure spikes and overheating.

Control Law Design

- Afterburner fuel schedule with feedforward + exit-temperature/thrust feedback; ignition sequencing and interlocks (pressure, flame sensors).
- Limiters to prevent upstream compressor or turbine damage.

Simulation & Test Cases

- A/B engage/disengage transients, re-ignition, failed ignition.
 - Metrics: ignition time, pressure spike magnitude, EGT/AFT limits.
-



Combustor — 1.3.3 Active Combustion Control (Thermo-Acoustic)

System Modeling

- Modal thermoacoustic dynamics: pressure oscillation modes coupled to heat release. Include actuator bandwidth (fuel modulation/acoustic driver).

Control Objectives

- Suppress thermoacoustic instabilities (reduce pressure oscillation amplitudes) across the operating envelope.

Control Law Design

- Band-limited feedback / adaptive notch controller or observer-based modal control. Robust/adaptive algorithms to cope with varying modal frequencies.
- Use a high-bandwidth actuator with phase compensation.

Simulation & Test Cases

- Inject disturbances at modal frequencies, vary operating point; verify suppression and no adverse effects.
- Metrics: RMS p' reduction, spectral peak attenuation, fuel modulation energy.

Combustor — 1.3.4 Air Inlet Control



System Modeling

- Inlet total pressure and distortion mapping; unsteady distortion dynamics and time delay to compressor face.

Control Objectives

- Maintain acceptable inlet conditions to preserve compressor surge margin and engine performance.

Control Law Design

- Command inlet guide/variable geometry (if present) and coordinate with compressor IGV/bleed. Gain-scheduled PI + feedforward from flight condition.
- Distortion detection with fallback modes.

Simulation & Test Cases

- Distorted inlet cases (asymmetric, parallel), gust encounter, rapid climb.
- Metrics: surge margin preservation, recovered pressure ratio, thrust loss.

Nozzle Swirler Complex (NSC) — 1.4.1 Swirl Number Control

System Modeling



- Vane geometry → swirl number → recirculation zone & mixing time constant. Model vane actuator dynamics.

Control Objectives

- Maintain swirl needed for flame anchoring and mixing; minimize pressure losses.

Control Law Design

- Lookup schedule for vane angle + PID trim using swirl proxy sensors (pressure pattern or chemiluminescence). Rate limits and deadband.

Simulation & Test Cases

- Sweep vane commands, fuel staging changes, monitor flame position and pressure drop.
- Metrics: swirl setpoint tracking, flame anchoring metrics, ΔP penalty.

Nozzle Swirler Complex (NSC) — 1.4.2 Flame Stability Control

System Modeling

- Equivalence ratio, local recirculation strength, and residence time coupling to flame blowout/flashback margin.

Control Objectives



- Avoid Lean Blowout (LBO) & flashback during throttle transients and altitude changes.

Control Law Design

- Supervisory logic that increases local fuel staging or swirl when LBO indicators approach threshold. Fast corrective inner loop on flame indicators. Fuzzy/supervisory control for large operating range.

Simulation & Test Cases

- LBO boundary tests, aggressive decel transients, fuel composition variation.
 - Metrics: LBO margin, recovery time, false-positive triggers.
-

Nozzle Swirler Complex (NSC) — 1.4.3 Emission Control

System Modeling

- Reduced NO_x/CO chemistry model (Zeldovich approximations or surrogate proxies) tied to peak flame temperature and residence time.

Control Objectives

- Minimize NO_x/CO while retaining performance and avoiding LBO or excessive T limits.

Control Law Design



- Constrained optimization / MPC at supervisory level adjusting staging, water injection, and equivalence ratio subject to Tt4 and thrust constraints. Schedule-based fallback.

Simulation & Test Cases

- Emission maps across operating points, tradeoff sweeps (NOx vs thrust vs efficiency).
 - Metrics: emission indices (NOx, CO), constraint violations, optimization convergence.
-

Fuel System — 1.5.1 Fuel Schedule Definition (Ndot)

System Modeling

- PLA→Ndot mapping including corrected speeds, ambient corrections; fuel flow dynamics and pressure sensitivity.

Control Objectives

- Provide safe Ndot demand that avoids surge, overtemp, and meets thrust commands.

Control Law Design

- Setpoint controller: ratio control (M.F / P3) + Ndot scheduling. Implement start/idle/max sequencing with limiters and interlocks. Adaptive scheduling for degradation.

Simulation & Test Cases



- Accelerate/decelerate profiles, start sequencing, degraded maps.
 - Metrics: adherence to Ndot profile, limiter activations, transient EGT response.
-

Fuel System — 1.5.2 Fuel Pumping & Metering Control

System Modeling

- Pump/regulator dynamics, flow orifice maps, pressure sensors and delays.

Control Objectives

- Accurate fuel flow delivery with stability despite supply pressure perturbations.

Control Law Design

- Cascade control: inner pressure/flow PI, outer fuel flow tracking controller with anti-windup; fault detection for leaks/pressure loss.

Simulation & Test Cases

- Supply pressure variation, valve stuck, sensor drift.
 - Metrics: flow tracking error, recovery time, stability under supply perturbations.
-



Fuel System — 1.5.3 Speed Governing with Accel/Decel Limiting

System Modeling

- Rotor inertia, torque coupling, $\dot{N}$ command $\rightarrow$ N response dynamics.

Control Objectives

- Enforce acceleration/deceleration limits for mechanical/thermal protection while meeting pilot demands.

Control Law Design

- Trajectory shaping ($\dot{N}$ limiter generator) with inner speed control loop; priority selection to honor T limits and surge margin. Adaptive limits for degraded engines.

Simulation & Test Cases

- Maximum allowable $\dot{N}$ commands, emergency throttle slam, repeated rapid cycles.
- Metrics: Not compliance, activations of limiters, impact on thrust response.

Fuel System — 1.5.4 Overall Gas-Generator Control Logic (FADEC Supervisor)

System Modeling

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- Integrated reduced-order model of compressor, combustor, turbine and fuel system with map-based components.

Control Objectives

- Coordinate subsystems, handle mode switching (start/idle/normal/afterburn/abnormal), and ensure safety limits.

Control Law Design

- Hierarchical FADEC: inner fast loops + Stateflow supervisory that handles scheduling, limit arbitration, fault management; optional supervisory MPC for optimized constrained operation.

Simulation & Test Cases

- Full mission profiles, degraded sensor/actuator cases, cross-coupled disturbances.
- Metrics: mission adherence, limit management success, fault handling behavior.

Fuel System — 1.5.5 Fuel Flow (Path & Priority)

System Modeling

- Entire fuel path, priority selector logic, metering valves dynamics, sensor network.

Control Objectives



- Accurate commanded fuel flow, safe priority selection during simultaneous limits (T limit vs surge margin), and health monitoring.

Control Law Design

- Feedforward demand + feedback cascade control; priority selector (min/max logic) between competing limiters; anti-windup and watchdogs for sensor faults.

Simulation & Test Cases

- Competing limiter scenarios (T vs surge), sensor failures, supply anomalies.

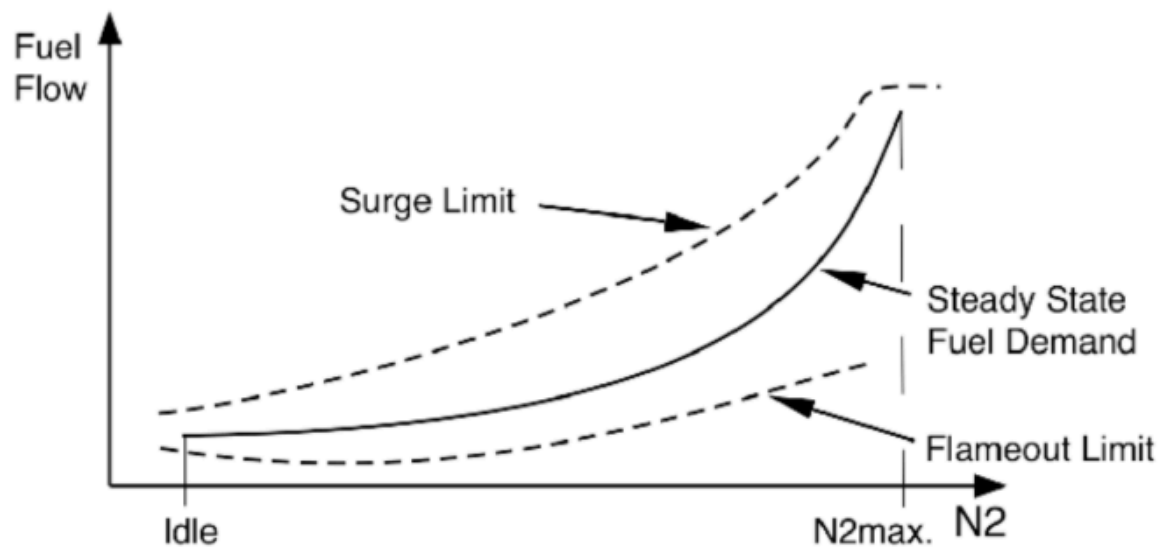
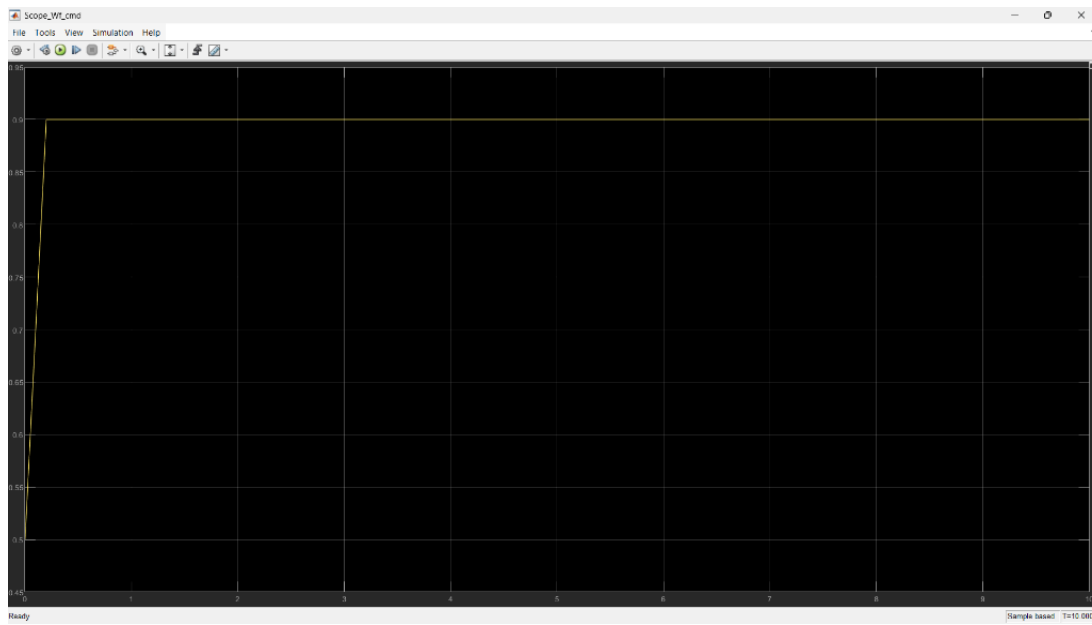


Fig. 5.1 Fuel demand curve of a turbine engine

- Metrics: correct priority enforcement, flow tracking, safe failover behavior.



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